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Mathematics, one idea at a time

This page is the 22-page reading map, written out. It is not a test. Stop when an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation before the idea has a sentence. The shelter, the truck, the shadow, and the coffee are teaching examples. They are not operating limits or engineering advice. The historical stories named in the sources are the ones with books behind them.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: say what you know, say what you want, and connect the two.

1. Algebra starts with an empty place

How many people in the shelter? Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An unknown is simply the answer you have not found yet. You can leave an empty box for it, or give it a letter as a label.

Here, let C stand for the total number of people. The letter is not a special object. It is the name on the empty place you want to fill.

The sign × means times, or multiplied by. The sign = means has the same value as. It is not an instruction to guess.

Three groups of forty people:

C = 3 × 40 = 120 people.

Check the answer. 40 people in the first room, 40 in the second, and 40 in the third make 120. The sign + means add: 40 + 40 + 40 = 120.

Writing C did not make the problem harder. It made the question visible: what total belongs here?

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements, and local rules.

1.2 Work backward, then check

How many pallets in the truck? A truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let P be the number of pallets. Before you know P, you can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

Name the unknown:

P × 500 pounds = 3,000 pounds.

"How many groups of 500 fit into 3,000?" is a division question. The sign ÷ means divided by. Division undoes multiplication.

Fill the empty place:

P = 3,000 ÷ 500 = 6 pallets.

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting the answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves P = 6.

Name the unknown, use the relationship, then check. The same habit returns with the ladder.

"lb" on a box means pounds. This is weight arithmetic only, not a loading plan. Real loads must meet vehicle, axle, space, balance, and securement limits.

2. Geometry gives the problem a shape

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE, about 2,300 years ago, he organized much earlier knowledge into a book called the *Elements*. BCE means before the common era. Geometry is much older than any one mathematician.

Geometry studies shapes, sizes, and positions. Euclid's lasting example was to begin with clear assumptions and build an explanation, step by step. A proof is reasoning that shows why a claim must be true.

An angle describes how widely two lines open apart. Angles are measured in degrees, written with the little sign °. A full turn is 360 degrees. A square corner is 90 degrees.

A right triangle has one square corner. "Right" here means 90 degrees, not the right-hand side of the page. Its hypotenuse is the side across from that corner, and it is always the longest side.

An equilateral triangle has three equal sides. On flat paper, each of its three angles is 60 degrees. It does not have a square corner.

These are different triangles. The equal-sided one can be split to make a right triangle. That split is why the ladder's one-half relationship is not a calculator accident.

2.2 Pythagoras connects three lengths

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A theorem is a statement established by proof. Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A square foot is a square one foot on each side.

The picture says: 3 × 3 + 4 × 4 = 5 × 5, or 9 + 16 = 25. The square on the 4-foot side is 16 square feet. The square on the 5-foot side is 25 square feet. If the long side were unknown, the question would be: what length times itself gives 25? The answer is 5.

The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning on the Moon. That calculation is in the ladder section below, and it is not the shadow shortcut.

2.3 Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.

In either half, the side opposite the 30-degree angle is 10 feet long. The sloping side, the hypotenuse, is 20 feet long. So the opposite side is half the hypotenuse.

Turn that half-triangle around on the page. The lengths and the angles do not change. It is the same shape as a 20-foot ladder reaching 10 feet up a wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not from a lucky measurement and not from a number invented by a calculator.

This ratio is called sine. Geometry supplies the shape. Trigonometry uses its proportions.

3. One ladder, three questions

Return to that right triangle. A ladder leaning against a wall makes the shape. The same ladder is kept throughout this section. Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown. Calculus is not required yet.

The drawing is on the ladder page, which keeps this same triangle. The numbers are repeated here so this walk stands on its own.

Right triangle: a triangle with one square corner, like the corner of a sheet of paper. Here that corner is where the wall meets the ground.

Hypotenuse, pronounced hy-POT-uh-noos: the side across from the square corner. It is always the longest side. Here it is the ladder.

Angle: how wide two lines open apart. Measure from the ground up to the ladder: 30 degrees. A square corner is 90 degrees.

This is a paper example, not advice for safely positioning a real ladder.

Sine, pronounced "sign"

How high does the ladder reach? The ladder is 20 feet long and makes a 30-degree angle with the ground. The wanted number is the height where it touches the wall.

Sine is the height divided by the ladder length. "Divided by" compares the two lengths. The answer is a ratio: how much of one length you have for each unit of the other.

For 30 degrees, the height is always half the ladder length. The equal-sided triangle above is why. A calculator or a table supplies the same ratio.

On a calculator, sin means sine, = means equals, and 0.5 means one half. So sin(30°) = 0.5.

Sine = height ÷ ladder length.

The height must be half of 20 feet:

20 feet × 0.5 = 10 feet.

Sine is the ratio, not the height itself. Multiplying by the ladder length gives the height.

Cosine, pronounced "co-sign"

How far is the foot from the wall? Keep the same ladder and the same ground angle. This time the wanted number is the base: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.

For 30 degrees, a calculator gives about 0.866. The base is about 0.866 times the ladder length. The short name is cos.

Cosine = base ÷ ladder length.

To find the base, multiply the ladder length by that ratio. The symbol ≈ means approximately equals, or about.

20 feet × 0.866 ≈ 17.32 feet.

"About" is required because 0.866 is a rounded calculator answer.

Relative to the ground angle, the height is the opposite side: it is across from the angle. The base is the adjacent side: it touches the angle and is not the hypotenuse.

Tangent, shortened to "tan"

What if the base is known and the ladder length is not? The height can still be found. Tangent compares the height directly with the base.

For a 30-degree angle, the tangent is about 0.577. For every foot along the ground, the triangle rises about 0.577 feet.

Tangent = height ÷ base.

A calculator writes tan(30°) ≈ 0.577. Here "tan" is the short name of the ratio. It is not another side and not another measurement.

17.32 feet × 0.577 ≈ 10 feet.

Why this works: a 40-foot ladder at 30 degrees reaches 20 feet high. Double the size. Keep the shape. The height-to-ladder ratio stays one half: 10 ÷ 20 = 20 ÷ 40 = 0.5.

All three ratios work this way. Keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

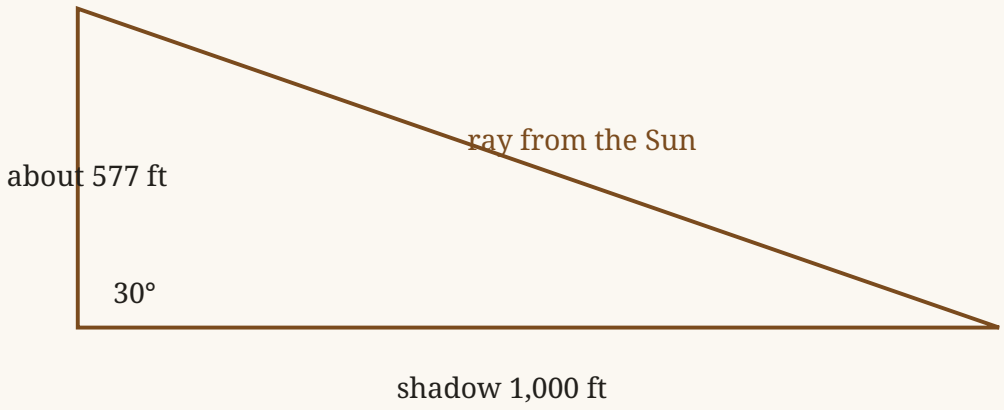
Calculator tip: select DEG, degrees, not RAD. A radian is another unit for the same opening. Use sin, cos, or tan with 30.

From a ladder to a lunar landscape

A shadow makes the same triangle. Replace the wall height with a mountain's height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The Sun's elevation angle means how high the Sun is above the local horizon, the level direction at that place. Use 30 degrees again.

Invented example, not Galileo's data



The upright line is the mountain's height. Level ground and one sun angle are assumed.

Mountain height about 577 feet from a 1,000-foot shadow at 30 degrees

Illustration only. The upright line is the mountain's height.

Tangent still means height divided by base.

Height = shadow length × tan(Sun angle).

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. The height is about 0.577 times the shadow length.

This is an invented example, not Galileo's data.

1,000 feet × 0.577 ≈ 577 feet.

This simple method also works on the Moon, approximately. You need the true ground length of the shadow, not an uncorrected length in a photograph, and the Sun's angle at that lunar place. The assumptions are level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature, and the viewing angle.

Galileo's Moon: what he actually did

The light reached the peaks first. In 1610 Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the terminator, the boundary between lunar day and night. He compared that distance with the Moon's diameter, its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used Pythagoras's rule, the relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation above. Both turn light and triangles into a height. They are different methods.

Galileo reported a height "more than 4 Italian miles." Those are historical Italian miles, not automatically four of today's English miles. This was his estimate, not a modern survey.

Take this one idea with you. An angle fixes the triangle's proportions. One known length then tells the scale. Each function answers a different question about the same ladder. Sine compares height with ladder length. Cosine compares base with ladder length. Tangent compares height with base. The names are new. The idea is comparing lengths.

Geometry gives a shape. Trigonometry compares its sides. If the shape or the position changes from moment to moment, that is calculus.

The shadow example is invented. Galileo's original text and the related shadow method are in the sources below.

4. Calculus: now let something change

The shelter total, the triangle, and the ladder all held still. A truck speeds up, a planet changes direction, and a tide rises. Calculus helps describe continuous change and what builds up from it.

In the late 1600s Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this mathematics closely with motion. It was not one person's invention out of nothing.

Two questions sit side by side. How fast, right now? That is a rate of change. How much altogether? That is accumulation.

A rate of change compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If the truck's speed is known throughout a journey, the small distances it covers can be added to find the whole trip.

These two questions are partners. From the journey's distance record, find the changing speed. From the changing speed, recover the distance traveled.

The earlier sections are foundations. They are not a claim that Pythagoras invented calculus. Geometry describes positions and lengths. Algebra expresses relationships. Calculus asks how those relationships behave as things change.

The central ideas can be understood before the formal symbols. The truck comes before the sky.

4.2 Average is not the same as right now

Suppose the truck travels 60 miles in two hours. Its average speed is the total distance divided by the total time: 60 ÷ 2 = 30 miles per hour. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment, the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.

On a graph, time runs across and distance runs upward. The curve records a journey. Its slope means rise compared with horizontal run: here, distance change compared with time change.

A straight line joining two points on the curve is the average over that interval. A tangent line follows the curve's direction at one point. Its slope gives the rate there. Finding that local rate is called differentiation. The rate found is called a derivative.

This is not the same use of "tangent" as tan on the ladder. There, tan is a triangle's height-to-base ratio. Here, a tangent line is a line beside a curve. Their connection is slope: rise divided by run.

The graph is conceptual. It is not the numerical record of the 60-mile trip above. No formula for the curve is required to read the idea.

4.3 Add the little pieces

Turn the question around. Suppose the speed is known. If the truck held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

The whole distance is 20 + 40 = 60 miles. A total was found by adding the contributions of different parts of the journey.

In a speed-versus-time graph, a rectangle's height is the speed and its width is the time. Multiplying them gives the distance during that time. The area under the speed graph therefore represents distance. The first hour is a rectangle 20 miles per hour high. The second hour is a rectangle 40 miles per hour high. Their areas are 20 miles and 40 miles.

When speed changes smoothly, divide the journey into short time pieces. Use each piece's speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called integration. The accumulated result is an integral.

Differentiation asks for the local rate. Integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 From moving trucks to moving planets

For Newton, motion raised a deeper question: how does a pull change an object's journey? Gravity is the attraction between masses, between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. Acceleration means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity, and acceleration from moment to moment.

Forward motion and the Sun's inward pull make the planet's path curve. Without gravity it would continue straight in its current direction. Orbits need not be circles. Many are ellipses, oval-shaped paths. Any sketch of that ellipse on this page's source sheet is schematic. Sizes and distances are not to scale.

The Moon pulls harder on Earth's near side than on its center, and less on the far side. This difference across Earth helps stretch the oceans and creates tides. The Sun contributes too.

Newton's gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape, and other effects. Calculus describes change. It does not make geography disappear.

Geometry describes shapes. Calculus describes how they change.

4.5 Why hot coffee cools more slowly

A fresh cup of coffee cools quickly at first, then more slowly as it approaches room temperature. A differential equation connects a changing quantity with its rate of change, how quickly it changes. Here is a simple model, called Newton's law of cooling.

First, name the pieces. T is the coffee's temperature, in degrees Fahrenheit. t is elapsed time, in minutes. T with the label room is the steady room temperature. "Room" is a label, not multiplication. The gap is coffee temperature minus room temperature: T − T_room. The sign − means subtract.

dT/dt reads "rate of temperature change": degrees Fahrenheit per minute. Think of dT as a tiny temperature change and dt as a tiny time change. The derivative describes the limiting rate. k is a positive cooling coefficient, a multiplier telling how quickly the gap shrinks. Its units are per minute, or inverse time.

The equals sign means both sides have the same value. Parentheses group the gap. Writing k beside parentheses means multiply. The equation reads:

dT/dt = −k(T − T_room).

In words: the size of the cooling rate is k times the gap. The minus sign before k makes the temperature-change rate negative. Hot coffee's temperature is falling. A smaller gap gives slower cooling.

The time constant is one characteristic cooling time. Its name is tau, written τ. The rule is τ = 1/k, one divided by k. Because k is per minute, tau is in minutes. The actual time depends on the cup and the conditions. No number of minutes has been chosen.

The sign % means out of a hundred. After one time constant, about 37% of the original gap remains: 63% has closed. After three, about 5% remains: 95% has closed. These percentages describe the gap, not the room temperature.

Example: coffee starts at 170°F in a 70°F room, a 100-degree gap. °F means degrees Fahrenheit. After one time constant it is about 107°F. After three, about 75°F. Check the first: 37% of 100 is 37, and 70 + 37 = 107.

The model assumes steady room temperature and an approximately constant k. It only approximates real coffee. It approaches room temperature without reaching it exactly in finite time. That is the local-rate idea at work.

4.6 The derivative, one instant at a time

On a coffee-temperature graph, time runs horizontally and temperature runs vertically. Choose an instant and mark its temperature as point A. Choose a nearby later instant and mark its temperature as point B. The letters are only labels.

Join the points. The line's slope is the vertical temperature change divided by the horizontal time change, rise divided by run. The run is positive. The rise is negative because coffee cools. The ratio is degrees Fahrenheit per minute.

Hold A fixed and slide B closer along the curve. Each two-point ratio estimates the slope at A. The value these ratios approach as the time gap shrinks is the limit: the derivative at A. Division by zero never happens.

Every instant has its own slope. In this cooling model the slopes become less negative as the curve flattens. The limit is exact for the ideal smooth model. Real measurements only estimate it.

The curve in the source sheet is schematic, not observed measurements. The line that joins A and B is the two-point estimate, not yet the derivative.

5. Six patterns, with the everyday examples

Input is the number you start with. Output is what the rule gives back. These tables use different units, so compare their patterns, not miles directly with square feet. The same tables are also on Six ways numbers grow. They are copied here so this paper is complete.

5.1 Linear: add the same amount

Linear growth adds the same amount for every equal step in the input. An odometer records 10 more miles each hour while a vehicle holds 10 miles per hour.

Driving time, hours	Trip distance, miles
1	10
2	20
3	30
4	40
5	50
6	60
7	70
8	80

5.2 Quadratic: the square pattern

Quadratic growth follows the area of a square as its side gets longer. A square garden bed 2 feet on each side covers 4 square feet. A square foot is a 1-foot square.

Side, feet	Area, square feet
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

5.3 Cubic: the cube pattern

Cubic growth follows the volume of a cube as its side gets longer. A concrete cube 2 feet on each side has 8 cubic feet of volume. A cubic foot is a 1-foot cube.

Side, feet	Volume, cubic feet
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512

5.4 Exponential: multiply each time

Exponential growth multiplies by the same factor at each equal step. Imagine bacteria doubling each hour, starting with one at hour zero. The table begins at hour 1.

Hours	Bacteria in the ideal model
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Bacteria cannot double forever. This model assumes unlimited resources. Doubling may start below a square pattern before eventually passing it.

5.5 Factorial: count possible orders

Factorial growth counts the different ways to place distinct people in a row of seats. Three different people can occupy a row of three seats in six different ways. n! , read "n factorial," multiplies the whole numbers from 1 through the group size n. The sign ! is not excitement.

People to seat, n	Seating arrangements
1	1
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320

5.6 Logarithmic: count multiplication steps

Logarithmic growth adds a fixed amount whenever the input grows by a fixed factor. log10 counts tenfold steps from 1. Each step multiplies by ten. The inputs are powers of ten, not the numbers 1 through 8. "Digits" counts written figures. 100 has three.

Input	Tenfold steps, log10	Written digits
1	0	1
10	1	2
100	2	3
1,000	3	4
10,000	4	5
100,000	5	6
1,000,000	6	7
10,000,000	7	8

Digit counts change in whole-number steps, not along a smooth log curve. "Log10 plus one" gives the digit counts shown for these powers of ten.

For these fixed rules, factorial eventually grows fastest. Small-input rankings differ. Compare patterns, not quantities with different units.

The twenty flight words, which use the same rate and the same integral on an imaginary trip, are Flight calculus, in plain language.

Follow the threads

The historical stories are real. The shelter, the truck, and the shadow numbers are teaching examples. This page reworks the standalone ladder lesson into its trigonometry section. That ladder lesson keeps its own page.

Geometry and its history: the University of St Andrews MacTutor histories of Euclid and Pythagoras. Euclid organized earlier knowledge. The right-triangle relationship predates Pythagoras.

Galileo's actual mountain calculation: *Sidereus Nuncius* (1610), E. S. Carlos translation (1880), pages 28 to 30, Project Gutenberg ebook 46036. Sunlit peaks beyond the terminator, a right-triangle calculation, "more than 4 Italian miles."

The related shadow method: Mary McIntyre, BBC Sky at Night Magazine, "How to measure craters on the Moon." Height equals shadow length times the tangent of the local Sun angle, with simplifying assumptions.

Calculus and its development: MacTutor, "A history of the calculus," for earlier work, Newton, Leibniz, rates, and accumulation. Orbits: NASA, "Gravity & Mechanics."

Why tides happen: NOAA, "Tides" and "What Causes Tides?" Uneven gravitational pull, the Moon and the Sun, and the influence of local geography.

Four questions, one growing toolkit. Algebra: what number is missing? Geometry: what shape and lengths are there? Trigonometry: what do the angles tell about the sides? Calculus: how is it changing, and what does it add up to? Recognizing the question is already a beginning.

Key facts

- Three rooms of 40 people: C = 3 × 40 = 120. Six pallets: P = 3,000 ÷ 500 = 6. Both answers are checked by putting them back.
- 3 × 3 + 4 × 4 = 5 × 5. An equilateral triangle of side 20 feet, split, shows why sine of 30 degrees is one half.
- A 20-foot ladder at 30 degrees reaches 10 feet and stands about 17.32 feet out. The 577-foot mountain is an invented shadow, not Galileo's "more than 4 Italian miles."
- Sixty miles in two hours is an average of 30 miles per hour. Area under a speed-time graph is distance: 20 + 40 = 60.
- Coffee at 170°F in a 70°F room has a 100-degree gap. After one time constant about 37 degrees of that gap remain, so the cup is about 107°F.
- The six patterns, including bacteria doubling and 8! = 40,320, are tabulated above.

The map, and not its earlier drafts, is Mathematics, one idea at a time (PDF).